

1. The wave-particle duality

de Broglie relationship

$$E=h\nu$$

$$p=h/\lambda$$

2. Ψ is a probability amplitude wave; $|\Psi|^2$

3. Measurement (average values)

4. Schrödinger equation

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi$$

$$\hat{H}\psi = E\psi$$

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2.1 Operators in quantum mechanics

An **operator** is a rule that transforms a given function into another function.

E.g. d/dx , $\sin$, $\log$

Eigenfunctions and Eigenvalues

Eigen is a German word meaning *characteristic*.

$$\hat{A}f(x) = kf(x)$$

Suppose that the effect of operating on some function $f(x)$ with the operator $\hat{A}$ is simply to multiply $f(x)$ by a certain constant k . We then say that $f(x)$ is an *eigenfunction* of $\hat{A}$ with *eigenvalue* k .

Operators obey the associative law of multiplication:

$$\hat{A}(\hat{B}\hat{C}) = (\hat{A}\hat{B})\hat{C}$$

A linear operator means

$$\hat{A}(\psi_1 + \psi_2) = \hat{A}\psi_1 + \hat{A}\psi_2$$

$$\hat{A}c\psi = c\hat{A}\psi$$

$$(\hat{A} + \hat{B})\psi \equiv \hat{A}\psi + \hat{B}\psi$$

$$(\hat{A} - \hat{B})\psi \equiv \hat{A}\psi - \hat{B}\psi$$

$$\hat{A}\hat{B}\psi \equiv \hat{A}[\hat{B}\psi]$$

Is $\frac{d}{dx}$ a linear operator

$$\left(\frac{d}{dx}\right)[f(x) + g(x)] = \frac{df}{dx} + \frac{dg}{dx} = \left(\frac{d}{dx}\right)f(x) + \left(\frac{d}{dx}\right)g(x)$$

$$\left(\frac{d}{dx}\right)[Cf(x)] = C \frac{df}{dx}$$

$\sqrt{\quad}$?

$$\sqrt{f(x) + g(x)} \neq \sqrt{f(x)} + \sqrt{g(x)}$$

Fundamental postulate of quantum mechanics (operator)

To every *physical property*, there corresponds a quantum-mechanical *Operator*.

$$q \rightarrow \hat{q} \qquad p_q \rightarrow \frac{\hbar}{i} \frac{\partial}{\partial q}$$

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} kx^2\right) \psi(x) = E \psi(x)$$

Eigenfunctions and eigenvalues

If the state function happens to be an eigenfunction of Q with eigenvalue q ,

$$\hat{Q}\psi_q = q\psi_q$$

Let's calculate the uncertainty ΔQ for this wavefunction. We need

$$\begin{aligned}\langle Q \rangle &= \frac{\int \psi_q^* \hat{Q} \psi_q d\tau}{\int \psi_q^* \psi_q d\tau} = \frac{\int \psi_q^* q \psi_q d\tau}{\int \psi_q^* \psi_q d\tau} = q \\ \langle Q^2 \rangle &= \frac{\int \psi_q^* \hat{Q}^2 \psi_q d\tau}{\int \psi_q^* \psi_q d\tau} = \frac{\int \psi_q^* q^2 \psi_q d\tau}{\int \psi_q^* \psi_q d\tau} = q^2\end{aligned}$$

(remember that q is just a number) and then

So *the uncertainty is zero*. This means that whenever we measure Q for a state with the wavefunction ψ_q , we *always* get the answer q .

$$(\Delta Q)^2 = \langle Q^2 \rangle - \langle Q \rangle^2 = q^2 - q^2 = 0$$

Prediction of measurements

Any normalized wavefunction ψ of the same variables and satisfying the same boundary conditions can be expressed in terms of normalized eigenfunctions ψ_k of $\hat{Q}$, so that

$$\psi = \sum_k c_k \psi_k$$

Here $\hat{Q}\psi_k = q_k\psi_k$, and the c_k are numerical coefficients, possibly complex. Then for a state with this wavefunction,

- If a measurement of Q yields the result q , then immediately after the measurement the system is in a state for which Q definitely has the value q . Consequently a measurement of Q will definitely yield one of the eigenvalues q_k .
- We cannot predict in advance which value will occur, but the probability of observing q_k is $|c_k|^2$.

Discussions for the Schrödinger equation

(c) Expectation Value

As an example, consider the expectation value of energy $\langle E \rangle$ for a discrete system is in state Ψ . The normalized eigenfunctions of energy are φ_n and the eigenvalues are E_n . The expectation (average) value is the sum:

$$\begin{aligned}\langle E \rangle &= \langle \Psi | \hat{H} | \Psi \rangle \\ &= \langle \sum_n b_n^* \varphi_n^* | \hat{H} | \sum_n b_n \varphi_n \rangle\end{aligned}$$

$$\langle E \rangle = \sum_{n=1}^{\infty} |b_n|^2 E_n$$

where $|b_n|^2$ is the absolute value of the coefficients of expansion that are given above. This is just the weighted average of the possible values of the observable E .

How to write Hamiltonian operators

For one particle, three dimensional system

$$\hat{H} = \hat{T} + \hat{V} = \frac{1}{2m} (p_x^2 + p_y^2 + p_z^2) + V(x, y, z)$$

$$\hat{H} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V(x, y, z) = -\frac{\hbar^2}{2m} \nabla^2 + V(x, y, z)$$

For three dimensional with n particles

$$\hat{T} = -\frac{\hbar^2}{2m_1} \left(\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial y_1^2} + \frac{\partial^2}{\partial z_1^2} \right) - \dots - \frac{\hbar^2}{2m_n} \left(\frac{\partial^2}{\partial x_n^2} + \frac{\partial^2}{\partial y_n^2} + \frac{\partial^2}{\partial z_n^2} \right) = -\sum_{i=1}^n \frac{\hbar^2}{2m_i} \nabla_i^2$$

$$\nabla_i^2 = \frac{\partial^2}{\partial x_i^2} + \frac{\partial^2}{\partial y_i^2} + \frac{\partial^2}{\partial z_i^2}$$

$$\hat{H} = -\sum \frac{\hbar^2}{2m_i} \nabla_i^2 + V(x_1, \dots, z_n)$$

$$\left[-\sum_{i=1}^n \frac{\hbar^2}{2m_i} \nabla_i^2 + V(x_1, \dots, z_n)\right] \psi = E\psi \quad \longleftarrow \quad \psi = \psi(x_1, y_1, z_1, \dots, x_n, y_n, z_n)$$

$$\left| \psi(x_1, y_1, z_1, x_2, y_2, z_2, \dots, x_n, y_n, z_n) \right|^2 dx_1 dy_1 dz_1 dx_2 dy_2 dz_2 \cdots dx_n dy_n dz_n$$

particle 1:

$$x_1 \rightarrow x_1 + dx_1, y_1 \rightarrow y_1 + dy_1, z_1 \rightarrow z_1 + dz_1$$

particle 2:

$$x_2 \rightarrow x_2 + dx_2, y_2 \rightarrow y_2 + dy_2, z_2 \rightarrow z_2 + dz_2$$

⋮

particle n:

$$x_n \rightarrow x_n + dx_n, y_n \rightarrow y_n + dy_n, z_n \rightarrow z_n + dz_n$$

$$\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \left| \Psi \right|^2 \underbrace{dx_1 dy_1 dz_1 \cdots dx_n dy_n dz_n}_{d\tau} = 1$$

$$\int \left| \Psi \right|^2 d\tau = 1$$

2.2 Hermiticity

Every operator $\hat{Q}$ has a *Hermitian conjugate*, conventionally denoted $\hat{Q}^\dagger$, which has the property that for any ψ_1 and ψ_2 satisfying the boundary conditions for the problem,

$$\int \psi_1^* \hat{Q} \psi_2 d\tau = \int (\hat{Q} \psi_1)^* \psi_2 d\tau$$

An operator that is equal to its Hermitian conjugate is said to be *Hermitian*.

Operators corresponding to physical observables must be Hermitian

$$\int \psi_1^* \hat{x} \psi_2 dx = \int (\hat{x} \psi_1)^* \psi_2 dx$$

so $\hat{x}^\dagger = \hat{x}$, and $\hat{x}$ is Hermitian.

$$\int \psi_1^* \frac{d}{dx} \psi_2 dx = \left[\psi_1^* \psi_2 \right]_{-\infty}^{\infty} - \int \left(\frac{d}{dx} \psi_1^* \right) \psi_2 dx = \int \left(-\frac{d}{dx} \psi_1 \right)^* \psi_2 dx \quad \text{So} \quad \left(\frac{d}{dx} \right)^\dagger = -\frac{d}{dx}$$

and d/dx is not a Hermitian operator. However id/dx is Hermitian

如果算符A等于其转置复共轭，或者说一个算符的转置复共轭等于自身，则称该算符为厄米算符。 $(A)^\dagger = A$

厄米算符既可以作用于内积的右侧项（右矢）也可以作用于左侧项（左矢），结果一样。

$$\hat{p}_x^\dagger = ?$$

$$\hat{p}_x^\dagger = \left(-i\hbar \frac{\partial}{\partial x} \right)^\dagger = i\hbar \left(\frac{\partial}{\partial x} \right)^\dagger = i\hbar \left(-\frac{\partial}{\partial x} \right) = -i\hbar \frac{\partial}{\partial x} = \hat{p}_x$$

Proof $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$

对于算符定域内任意波函数 φ, ϕ

$$\langle \varphi | (\hat{A}\hat{B})^\dagger | \phi \rangle = \langle \hat{A}\hat{B} | \varphi \phi \rangle = \langle \hat{B}\varphi | \hat{A}^\dagger | \phi \rangle = \langle \varphi | \hat{B}^\dagger \hat{A}^\dagger | \phi \rangle$$

Dirac notation

We sometimes use a notation due originally to Dirac. The idea is to reduce notational clutter and give more prominence to the labels identifying the wavefunctions.

In this notation $|n\rangle$ is used for the wavefunction ψ_n . $|n\rangle$ is called a *ket*.

$\langle n|$ is a *bra*. The bra notation implies the complex conjugate ψ_n^* .

A complete *bracket* expression, like $\langle n|n\rangle$ or $\langle n|\hat{Q}|n\rangle$, implies integration over all space.

Thus the notation $\langle n|n\rangle$ means the integral $\int \psi_n^* \psi_n d\tau$, and $\langle m|\hat{Q}|n\rangle$ means $\int \psi_m^* \hat{Q} \psi_n d\tau$.

Using this notation, the expectation value integral can be written more compactly as

$$\langle Q \rangle = \frac{\int \psi^* \hat{Q} \psi d\tau}{\int \psi^* \psi d\tau} \quad \Rightarrow \quad \langle Q \rangle = \frac{\langle n|\hat{Q}|n\rangle}{\langle n|n\rangle}$$

Hermiticity

Properties of Hermitian operators:

- Their eigenvalues are always real.
- Eigenfunctions corresponding to different eigenvalues are orthogonal.

In the proof of these properties we use Dirac's angle-bracket notation. First note that if $\hat{Q}$ is Hermitian, then

$$\begin{aligned}\langle m|\hat{Q}|n\rangle^* &\equiv \left(\int \psi_m^* \hat{Q} \psi_n d\tau\right)^* \\ &= \left(\int (\hat{Q} \psi_m)^* \psi_n d\tau\right)^* \\ &= \int \psi_n^* \hat{Q} \psi_m d\tau \\ &= \langle n|\hat{Q}|m\rangle\end{aligned}$$

Note also that $\langle m|n\rangle^* = \left(\int \psi_m^* \psi_n d\tau\right)^* = \int \psi_n^* \psi_m d\tau = \langle n|m\rangle$.

Hermiticity and orthogonality

Now consider two eigenfunctions $|m\rangle$ and $|n\rangle$. We have

$$\hat{Q}|m\rangle = q_m |m\rangle$$

$$\langle n|\hat{Q}|m\rangle = q_m \langle n|m\rangle$$

$$\langle m|\hat{Q}|n\rangle = q_m^* \langle m|n\rangle$$

and

$$\hat{Q}|n\rangle = q_n |n\rangle$$

$$\langle m|\hat{Q}|n\rangle = q_n \langle m|n\rangle$$

where the last line on the left comes from taking the complex conjugate.

Subtracting, we find

$$0 = (q_n - q_m^*) \langle m|n\rangle$$

and from this we can deduce

- (a) If $m = n$, then $\langle m|n \rangle = \langle m|m \rangle \neq 0$, so $q_m = q_m^*$ and q_m is real.
- (b) If $q_m \neq q_n$, then since both are real, $q_n - q_m^* \neq 0$ and $\langle m|n \rangle = 0$.

Hermitian operator ensures that the eigenvalue of the operator is a real number

- **The eigenvalue of a Hermitian operator is a real number**
- **The eigenfunctions of Hermitian operators are orthogonal**

量子力学算符必须是厄米算符

1. 物理量的测量结果必须是实数，一个物理量 A ，它的期望值 $\langle A \rangle$ 是在大量重复实验中测量结果的平均值。这个平均值显然也必须是一个实数。
2. 厄米算符属于不同本征值的本征态是相互正交的，并且所有这些本征态共同构成一个完备集，可以作为希尔伯特空间的一组基。这意味着系统的任意一个量子态 $|\psi\rangle$ 都可以用这些本征态展开

厄米算符：

- (1) 厄米算符乘以常数，仍然是厄米算符；
- (2) 多个厄米算符的线性组合仍然是厄米算符；
- (3) 两个厄米算符的乘积呢？

2.3 Expansion in terms of eigenfunctions

Postulate: If $\psi_1, \psi_2, \dots, \psi_n$ are the possible states of a microscopic system, then the linear combination of these states is also a possible state of the system.

$$\Psi = c_1\psi_1 + c_2\psi_2 + c_3\psi_3 \cdots + c_n\psi_n = \sum_i c_i\psi_i$$

We now postulate that the set of eigenfunctions of any Hermitonian Operator that represent a physical quantity forms a complete set.

2.4 Commutation

When two or more operators are applied to a wavefunction, the order matters.

$\hat{P}\hat{Q}\psi$ is defined to mean $\hat{P}(\hat{Q}\psi)$, and it may be different from $\hat{Q}\hat{P}\psi \equiv \hat{Q}(\hat{P}\psi)$.

Example: For any $\psi(x)$,

$$\hat{x}\hat{p}_x\psi = -i\hbar x \frac{d\psi}{dx}$$

but

$$\hat{p}_x\hat{x}\psi = -i\hbar \frac{d}{dx}(x\psi) = -i\hbar \left(\psi + x \frac{d\psi}{dx} \right)$$

Subtracting the second of these from the first:

$$(\hat{x}\hat{p}_x - \hat{p}_x\hat{x})\psi = i\hbar\psi \quad \longrightarrow \quad \hat{x}\hat{p}_x - \hat{p}_x\hat{x} = i\hbar$$

This is true for any wavefunction ψ , so we can extract the *operator identity*

$$[\hat{3}, \frac{d}{dx}] = 3 \frac{d}{dx} - \frac{d}{dx} 3$$

$$(3 \frac{d}{dx}) f(x) = 3 \frac{df}{dx} \quad , \quad (\frac{d}{dx} 3) f(x) = \frac{d}{dx} [3 f(x)] = 3 \frac{df}{dx}$$

$$[\hat{3}, \frac{d}{dx}] = 0$$

$$[\frac{d}{dx}, x] = \frac{d}{dx} \cdot x - x \frac{d}{dx}$$

$$(\frac{d}{dx} \cdot x) f(x) = f(x) + x \frac{df}{dx}$$

$$(x \frac{d}{dx}) f(x) = x \frac{df}{dx}$$

$$[\frac{d}{dx}, x] = 1$$

We say that $\hat{x}$ and $\hat{p}_x$ do not commute.

The expression $\hat{x}\hat{p}_x - \hat{p}_x\hat{x}$ is called the *commutator* of $\hat{x}$ and $\hat{p}_x$, and there is a special notation for it: $[\hat{x}, \hat{p}_x]$.

If $\hat{P}\hat{Q} = \hat{Q}\hat{P}$, $\hat{P}$ and $\hat{Q}$ are said to *commute*. A constant commutes with any operator: $[k, \hat{P}] = 0$.

The uncertainties in two observables P and Q satisfy

$$\Delta P \Delta Q \geq \frac{1}{2} \left| \langle [\hat{P}, \hat{Q}] \rangle \right|$$

This is *Heisenberg's uncertainty principle* in a more general form than before.

Since $[\hat{x}, \hat{p}_x] = i\hbar$, we see that $\Delta x \Delta p_x \geq \frac{1}{2} \hbar$,

不确定关系就是两个不能对易的力学量测量的标准误差偏移大小的关系

当状态不是该力学量的本征函数时，测出来的力学量是平均值。每个力学量测定的数值和平均值偏差多大？

不确定原理是微观粒子运动的基本规律，它是“波粒二象性”和“波函数统计解释”导致的必然结果（严格上不算是一个独立的原理）。我们常常把它称作“不确定关系”或“测不准关系”，更多的是表达不对易的两个算符（没有共同本征函数）所对应物理量同时测量结果的误差关系，因为它们的测量手段本身不同，甚至在同一时刻下是互斥的，客观上导致它们不能被同时准确测量。

Commutation and eigenfunctions

If $\hat{P}$ and $\hat{Q}$ commute, i.e., $[\hat{P}, \hat{Q}] = 0$, it is possible for both ΔP and ΔQ to be zero — that is, we can **find wavefunctions that are eigenfunctions of both**.

In fact, in this case, a non-degenerate eigenfunction of $\hat{P}$ must also be an eigenfunction of $\hat{Q}$.

Proof: Suppose that
$$\hat{P}\psi = p\psi$$

and that there is no other wavefunction with this eigenvalue for $\hat{P}$ (except for numerical multiples of ψ , which are essentially the same).

Now consider $\hat{Q}\psi$. $\hat{P}$ and $\hat{Q}$ commute, so

$$\hat{P}(\hat{Q}\psi) = \hat{Q}\hat{P}\psi = \hat{Q}p\psi = p(\hat{Q}\psi)$$

So $(\hat{Q}\psi)$ is an eigenfunction of $\hat{P}$ with eigenvalue p .

But ψ , or a multiple of it, is the only such function. Therefore $\hat{Q}\psi = q\psi$, for some number q ; i.e., ψ is an eigenfunction of $\hat{Q}$.

A necessary condition for the existence of a complete set of simultaneous eigenfunctions of two operators is that **the operators commute with each other**.

Conversely, if $\hat{P}$ and $\hat{Q}$ are two commuting operators, there exists a complete set of functions that are eigenfunctions of both $\hat{P}$ and $\hat{Q}$.

$$[A, B + C] = [A, B] + [A, C]$$

$$[A, BC] = B[A, C] + [A, B]C$$

$$[AB, C] = A[B, C] + [A, C]B$$

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$$

Example:

$$[\hat{x}, \hat{p}_x^2] = [\hat{x}, \hat{p}_x] \hat{p}_x + \hat{p}_x [\hat{x}, \hat{p}_x] = i\hbar \frac{\hbar}{i} \frac{\partial}{\partial x} + \frac{\hbar}{i} \frac{\partial}{\partial x} i\hbar = 2\hbar^2 \frac{\partial}{\partial x}$$

$$[\hat{x}, \hat{H}] = [\hat{x}, \hat{T} + \hat{V}] = [\hat{x}, \hat{T}] + [\hat{x}, V(x, y, z)] = [\hat{x}, \hat{T}]$$

$$= [\hat{x}, \frac{1}{2m} (p_x^2 + p_y^2 + p_z^2)]$$

$$= \frac{\hbar^2}{m} \frac{\partial}{\partial x}$$

$$= \frac{i\hbar}{m} \hat{p}_x$$

We could not predict the position of particle for a stationary state.

Time evolution of expectation

$$\bar{F} = \langle \Psi | \hat{F} | \Psi \rangle$$

$$\begin{aligned} \frac{d\bar{F}}{dt} &= \left\langle \frac{\partial \Psi}{\partial t} \right| \hat{F} | \Psi \rangle + \langle \Psi | \hat{F} \left| \frac{\partial \Psi}{\partial t} \right\rangle + \langle \Psi | \frac{\partial \hat{F}}{\partial t} | \Psi \rangle \\ &= -\frac{1}{i\hbar} \langle \hat{H} \Psi | \hat{F} | \Psi \rangle + \frac{1}{i\hbar} \langle \Psi | \hat{F} \hat{H} | \Psi \rangle + \frac{\partial \hat{F}}{\partial t} \\ &= -\frac{1}{i\hbar} \langle \Psi | \hat{H} \hat{F} | \Psi \rangle + \frac{1}{i\hbar} \langle \Psi | \hat{F} \hat{H} | \Psi \rangle + \frac{\partial \hat{F}}{\partial t} \\ &= \frac{1}{i\hbar} [\hat{F}, \hat{H}] + \frac{\partial \hat{F}}{\partial t} \end{aligned}$$

$$-i\hbar \frac{d\psi^*}{dt} = H\psi^* \quad i\hbar \frac{d\psi}{dt} = H\psi$$

$$\frac{d\bar{F}}{dt} = \frac{\partial \hat{F}}{\partial t} + \frac{1}{i\hbar} [\hat{F}, \hat{H}]$$

$$\text{If} \quad \partial \hat{F} / \partial t = 0 \quad \frac{d\bar{F}}{dt} = \frac{1}{i\hbar} \overline{[\hat{F}, \hat{H}]}$$

$$\text{If } F \text{ commutes with } \hat{H}, \quad [\hat{F}, \hat{H}] = 0 \quad d\bar{F}/dt = 0$$

$\mathbf{F}$ and $\hat{H}$ have a common complete set of eigen wavefunction. Therefore, when a time-independent operator commutes with H , it is a constant of motion.

$$\frac{d\bar{H}}{dt} = \overline{\frac{\partial \hat{H}}{\partial t}} + \frac{1}{i\hbar} \overline{[\hat{H}, \hat{H}]} = 0$$

厄米算符性质：

- (1) 在任何状态下，厄米算符的本征值必为实数；
- (2) 在任何状态下平均值为实数的算符必为厄米算符；
- (3) 厄米算符的属于不同本征值的本征函数彼此正交；
- (4) 厄米算符的本征函数具有完备性。

量子体系中的可观测量（力学量）用线性厄米算符来描述：

- 算符的线性是状态叠加原理所要求的
- 实验上的可观测的力学量总是实数，力学量相应的算符必须是厄米算符
- 力学量之间的关系可通过相应算符之间的对易关系来反映出来